

求极限：

$$\lim_{x \rightarrow 0} \int_0^x \frac{\int_t^x e^{-(u-t)^2} du}{1 - e^{-x^2}} dt$$

首先看分子积分：

$$\int_0^x \int_t^x e^{-(u-t)^2} du dt$$

令 $a = u - t, b = u + t$ 则 $du dt = \frac{1}{2} da db$ ，则积分可以化为：

$$\begin{aligned} & \frac{1}{2} \int_0^x \int_a^{2x-a} e^{-a^2} db da \\ &= \int_0^x (x - a) e^{-a^2} da \\ &= x \int_0^x e^{-a^2} da - \frac{1}{2} (1 - e^{-x^2}) \end{aligned}$$

又 $e^{-x^2} = 1 - x^2 + o(x^2)$

因此

$$\begin{aligned} & \lim_{x \rightarrow 0} \int_0^x \frac{\int_t^x e^{-(u-t)^2} du}{1 - e^{-x^2}} dt \\ &= \lim_{x \rightarrow 0} \frac{x \int_0^x 1 - a^2 da}{1 - (1 - x^2)} - \frac{1}{2} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{x^2} - \frac{1}{2} = \frac{1}{2} \end{aligned}$$